CHEAT SHEET UTS · WEEKS 1–8

MK 4 · Operations & Supply Chain Management

Don't mix these up

- Efficiency = low cost; effectiveness = right value. Qualifier = to be considered; winner = to be chosen.
- ERRC: Eliminate & Reduce cut cost; Raise & Create add value.
- Slack 0 = critical path. Crash the cheapest critical activity first. CPI = EV/AC, SPI = EV/PV (<1 = bad).
- Utilisation = used ÷ best operating level. Services target ~70%. Add capacity at the bottleneck.
- Queues come from variability. ρ → 1 makes Lq explode. Constant service halves the queue.

W1 Introduction to Operations & Supply Chain Management

What is OSCM? · The five supply chain processes · How services differ from goods · History, sustainability and analytics · Efficiency, effectiveness, value and efficiency ratios · Southwest in numbers: ground minutes = 'free' aircraft · Honda's efficiency ratios charted

- OSCM = design, operation, improvement of processes.
- Sketch the network first.
- Plan, source, make, deliver, return.
- Services: intangible, interactive, heterogeneous, perishable, package.
- Efficiency ≠ effectiveness.
- Asset turnover is the most comprehensive ratio; compare within industry.

W2 Operations Strategy & Blue Ocean Strategy

Operations strategy and sustainability · Competitive dimensions, qualifiers and winners · IKEA's activity system, risk and productivity · Blue Ocean: value innovation · Blue Ocean tools: canvas, four actions, ERRC · Competitive trade-offs: IKEA vs a traditional furniture store · The productivity paradox: partial up, total down · [yellow tail]'s strategy canvas: a different shape

- Operations strategy guides resource use; trade-offs are unavoidable.
- Qualifiers vs winners — winners migrate.
- IKEA wins through activity fit.
- Partial productivity can hide falling total productivity.
- Blue ocean: differentiation and low cost via ERRC.
- Focus, divergence, tagline.

FORMULAS

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Productivity = Output ÷ Input
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W3 Design of Products and Services

Why design is an operations problem · Development process: six phases and seven variants · Design criteria: QFD, DFMA, VA/VE, ecodesign · Service design and new-product economics · Design Thinking and SCAMPER · A simple House of Quality: the car door · New-product NPV sensitivity · Apple and Shell in numbers

- Design locks in cost and capability needs.
- Six phases, seven variants.
- QFD brings the customer's voice in.
- DFMA, VA/VE, ecodesign.
- Complexity and divergence are design choices.
- NPV plus sensitivity.
- Design Thinking loop; SCAMPER provocations.

W4 Project Management

Projects, structures and the triple constraint · CPM: critical path and slack · Three time estimates (PERT) · Compressing: fast tracking vs crashing · Earned Value Management · Choosing the delivery approach · The sample network as a Gantt: critical path and slack made visible · The time-cost curve: why fastest ≠ cheapest · Earned-value S-curves: reading a project in month 4 · Whoosh in charts

- State confidence, not just dates.
- The triple constraint relocates pressure.
- Critical path; slack = LS − ES.
- TE = (a+4m+b)/6.
- Crash the cheapest critical activity.
- CPI = EV/AC, SPI = EV/PV.
- Set kill criteria before investing.

FORMULAS

```
Slack = LS − ES = LF − EF
```

```
TE = (a + 4m + b) ÷ 6    σ² = ((b − a) ÷ 6)²
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Biaya percepatan per periode = (crash cost − normal cost) ÷ (normal time − crash time)
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```
CV = EV − AC; CPI = EV ÷ AC    SV = EV − PV; SPI = EV ÷ PV
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```
EAC ≈ BAC ÷ CPI = 100 ÷ 0.87 ≈ 115
```

W5 Strategic Capacity Planning

Capacity as a strategic commitment · Scale, focus and flexibility · Planning capacity: cushion, chunk, balance · Decision trees for capacity alternatives · Service capacity is different · RBV/VRIN and dynamic capabilities · The U-shaped unit-cost curve and best operating level · Hackers' decision tree: before and after discounting · Why services target ~70%: queues explode at high utilisation · Semarang terminal in charts

- Capacity is a rate.
- Utilisation = used ÷ BOL.
- The cushion is a decision.
- Add at the bottleneck.
- Solve trees right to left.
- Services target ~70%.
- Transform before you duplicate.

FORMULAS

```
Capacity utilization rate = Capacity used ÷ Best operating level
```

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Capacity cushion = (Kapasitas − Permintaan) ÷ Permintaan
```

W6 Process Design and Analysis

Processes and the decoupling point · Layouts, product–process matrix, break-even · Customer contact and the service-system design matrix · BPMN 2.0 and the service blueprint · Break-even chart for equipment choice · The customer-contact continuum: efficiency vs sales opportunity · KAI capacity: why a one-second gate doesn't fix the queue

- Decoupling point trades speed for inventory cost.
- Stay on the diagonal.
- Break-even equipment choice.
- Contact lowers efficiency, raises sales chance.
- Accommodate or reduce variability.
- Blueprint finds, BPMN fixes.
- The slowest resource sets capacity.

FORMULAS

```
Titik B: 80.000 ÷ (200 − 75) = 640 unit
Titik A: (200.000 − 80.000) ÷ (75 − 15) = 2.000 unit
```

```
Ceiling = 80 ÷ share needing registration
```

W7 Waiting Lines and Simulation

The waiting line problem and variability · Anatomy of a queuing system · The four models and formulas · When formulas fail: simulation · Queue simulator: feel the explosion yourself · Variability: constant service halves the queue · Bob & Ray and Manggarai: when formulas aren't enough

- Variability creates queues.
- Constant service halves the queue.
- $Lq = \lambda^2/[\mu(\mu-\lambda)]$.
- $L = \lambda W$.
- Queues explode as $\rho \to 1$.
- Averages hide queues — use P95.
- Escalate modelling only when forced.

FORMULAS

```
Aproksimasi dua momen (Kingman): Wq ≈ [ρ ÷ (1 − ρ)] × [(Ca² + Cs²) ÷ 2] × waktu layanan
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L = λ × W
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Kingman: Wq ≈ [ρ/(1−ρ)] × [(Ca² + Cs²)/2] × service time
```

W8 Lean Operations & Supply Chain

Learning objectives and roadmap · Lean starts with value, not cost · The seven wastes · Toyota Production System: five design choices and their preconditions · The Toyota house · Sizing a kanban loop · Inventory hides problems — and what lean assumes · Lean tools: what each buys and what it costs · From lean plant to lean supply chain · Lean in services · The Lean Startup: Build–Measure–Learn · What an MVP is and isn't — and when the Lean Startup fits · Six takeaways

- Lean = removing waste; value = what the customer will pay for (defined outside); waste is seen from inside the process.
- Typically <10% of elapsed time adds value; 90%+ is waiting, moving, reworking, sitting.
- Seven wastes: overproduction (root), inventory, waiting, transport, motion, over-processing, defects.
- TPS = five design choices — pull, kanban, small lots (setup first), heijunka, quality at source — each with a precondition.
- Toyota house: goal on the roof, JIT and Jidoka pillars, stability and culture foundations.
- Kanban: $k = D \times L \times (1 + S) \div C$, round up; attack L, not k.
- Inventory hides problems; removing buffers without removing variability just moves the problem.
- Lean chain: suppliers, procurement, warehousing, logistics, customers; outside your walls change must be negotiated.
- Lean services attack time, motion and variation; lean removes waste, not reassurance.
- Lean Startup: waste = anything producing no validated learning; Build–Measure–Learn; MVP = least thing that could disprove the idea; persevere or pivot.

FORMULAS

```
k = D × L × (1 + S) ÷ C
```